RETAIL SALES PERFORMANCE ANALYSIS

BUSINESS REPORT

Project 1 | Tool: Python | Dashboard: Streamlit

1. Executive Summary

This project analyzes retail sales data to understand sales performance across regions, product categories, customer segments, products, time periods, and discount levels. The analysis combines data cleaning, exploratory analysis, visualization, and an interactive Python dashboard to support data-driven business decisions.

2. Business Problem

The retail company has experienced inconsistent sales performance across different regions. Management needs to identify which products, locations, and customer segments are driving revenue and which areas require improvement.

3. Objectives

- Analyze overall sales and profit performance.
- Compare sales performance across regions.
- Identify high-performing and low-performing product categories.
- Understand customer segment contribution to sales.
- Analyze sales trends over time.
- Identify top products by sales and profit.
- Investigate the relationship between discount levels and profit.
- Provide practical recommendations for management.

4. Dataset Description

The Retail_Sales.csv dataset contains 9,994 retail transaction records and 21 fields. Important fields include Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Country, City, State, Region,

Product ID, Category, Sub-Category, Product Name, Sales, Quantity, Discount, and Profit.

5. Data Preparation

- Loaded the CSV file using pandas with latin1 encoding.
- Converted Order Date and Ship Date into datetime format.
- Checked missing values and duplicate records.
- Reviewed numerical summary statistics and categorical values.
- Created Year and Month fields for time-based analysis.
- Calculated business KPIs including sales, profit, quantity, orders, customers, and average order value.

6. Key Analysis Areas

Analysis	Business Question	Decision Use
Region	Which regions generate the most sales?	Regional investment and targeting
Category	Which categories drive sales and profit?	Product and inventory strategy
Segment	Which customer segments contribute most?	Customer targeting
Time Trend	How do sales change over time?	Planning and seasonality
Products	Which products perform best?	Product prioritization
Discount	How does discounting affect profit?	Pricing and promotion control

7. Business Insights

- Sales performance should be compared with profit rather than revenue alone, because high sales do not always mean high profitability.
- Regional differences can reveal locations where sales are strong and locations where additional marketing or operational improvement may be needed.
- Category and product analysis helps management focus inventory and promotional activity on products that contribute strongly to business performance.
- Customer segment analysis helps identify the customer groups that generate the greatest revenue and can guide targeted campaigns.
- The monthly trend provides visibility into growth, declines, and possible seasonal patterns.
- The discount analysis is important because aggressive discounts can increase sales while reducing profit margins.

8. Recommendations

1. Monitor profit together with sales when evaluating products and promotions.
2. Prioritize high-performing regions and investigate weak regions for operational or marketing issues.
3. Maintain sufficient stock for strong-selling products and review consistently weak products.
4. Use customer segment information to create targeted promotions instead of applying the same offer to every customer.
5. Review high-discount transactions carefully and set discount limits where profitability is negatively affected.
6. Use the interactive dashboard regularly to monitor changes and support management decisions.

9. Dashboard

The Streamlit dashboard provides interactive filters for Region, Category, and Customer Segment. It displays KPI cards, sales by region, sales and

profit by category, sales by segment, monthly sales trend, top products, discount versus average profit, and a filtered data preview.

10. Conclusion

The project demonstrates how Python can transform raw retail transaction data into useful business information. The combination of pandas for analysis, matplotlib for visualization, and Streamlit for interactive reporting provides management with a practical way to monitor performance and identify areas for improvement.

11. Deliverables Checklist

- Retail_Sales.csv dataset
- Python analysis script (analysis.py)
- Interactive dashboard (dashboard.py)
- Analysis charts/screenshots
- Business report